

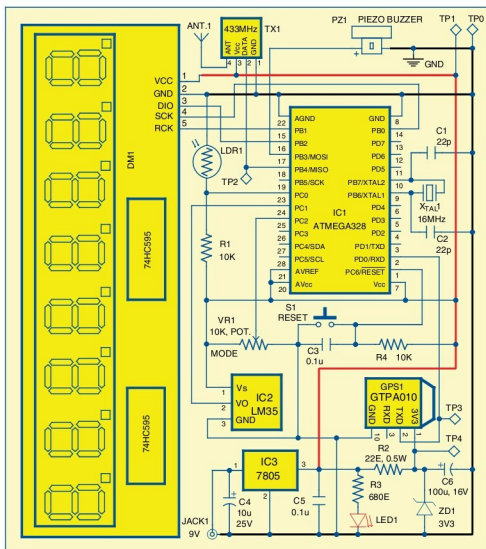


Fig. 3: Circuit diagram of the GPS master clock



Fig. 4: 8-digit, 7-segment display shield for Arduino

the time. The master clock receives GPS signals from satellites and so the time (that is, coordinated universal time, or UTC time) is precise all the time. One just has to set the user country time from the UTC time. The slave clock receives regular signals from the master clock through a pair of 433MHz RF modules. These clocks always pick time from GPS signals,

and slave clock circuits.

GPS master clock. The circuit diagram of the GPS master clock is shown in Fig. 3.

ATmega328 microcontroller. The heart of the master clock circuit is ATmega328 microcontroller, which is programmed with the help of Arduino. ATmega328 has 32 kilobytes of flash memory for the whole work

to be carried out. ATmega328 chip is normally found in an Arduino board. If you want to get a fresh chip from the market, you should opt for a chip that has the Arduino boot loader.

Display module (DM1). The 7-segment display used here is an 8-digit, Arduino shield as shown in Fig. 4. The shield consists of eight common-anode 7-segment displays (12.7mm or ½-inch), multiplexed on 16 output pins of two inexpensive 74HC595 shift registers.

Apart from displaying time and date, the project can also display other information like latitude, longitude, altitude, speed, temperature and light intensity. All this information will be displayed on the display module one by one, continuously in a cyclic manner.

To control the information display, we have used analogue pin 25 (PC2) of IC1. Digital values vary from 0 to 1023 to get different information on the 7-segment display using a 10k potmeter. To display only one piece of information, say, temperature, vary VR1 slowly until you can see a constant

so time setting or reset is never required.

Circuit and working

The complete project includes GPS master clock

temperature data on the display.

For temperature and light, we have used two inexpensive sensors LM35 (IC2) and LDR (LDR1), besides the GPS receiver module (GPS1). Also, we have added one small buzzer (PZ1) to sound an alarm every 15 minutes between 5am and 11pm.

Since a 7-segment display has limitations with respect to displaying all characters, we have used a suitable format in a readable way, like, tinE = time, lat. = latitude, lon. = longitude, alt. = altitude, SPD = speed, date = date, lit. = light and the like. To display, for example, the latitude, longitude and altitude of the current position of the GPS master clock, vary VR1 till you get lat., lon. and alt. along